UNIT-IV

MATRIX DECOMPOSITION

Diagonalization Of Matrix:

→ Let A be a square matrix of order n, let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigen values then $x_1, x_2, \ldots, x_n$ are the corresponding eigen vectors. Then we have $Ax_i = \lambda_i x_i \; \forall i$ where $x_1, x_2, \ldots; x_n$ are linearly independent. Construct a matrix P called diagonalization matrix or model matrix $P = [x_1, x_2, \ldots x_n]$.

Consider $AP = A[x_1, x_2, \ldots, x_n]$

$= [Ax_1, Ax_2, \ldots, Ax_n]$

$= [\lambda_1 x_1, \lambda_2 x_2, \ldots, \lambda_n x_n]$

$= [x_1, x_2, \ldots, x_n] \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & & \\ 0 & & \lambda_3 & \\ 0 & & & \lambda_n \end{bmatrix}$

$AP = PD$

$P^{-1}AP = D$

$A = PDP^{-1}$

where D- Diagonal matrix with eigen values as a diagonal elements.

I. Diagonalize the matrix $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ Hence find A^6

⇒ $A - \lambda I$

NOTE

→ For any +ve integer n we can f

$P D^n P^{-1}$.

$\begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 7-\lambda & 2 \\ -4 & 1-\lambda \end{bmatrix}$

$|A - \lambda I| = 0$

$\begin{vmatrix} 7-\lambda & 2 \\ -4 & 1-\lambda \end{vmatrix} = 0$

$\lambda^2 - tr(A) + |A| = 0$

$\lambda^2 - 8\lambda + 15 = 0$

$(\lambda - 5)(\lambda - 3) = 0$

$\lambda = 5, 3$

To find eigen vectors,

$(A - \lambda I)x = 0$

when $\lambda = 5$,

$\begin{bmatrix} 7-5 & 2 \\ -4 & 1-5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$

$\begin{bmatrix} 2 & 2 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$

$2x_1 + 2x_2 = 0$

$-4x_1 - 4x_2 = 0$

NOTE

→ For any +ve integer n we can find out $A^n = P D^n P^{-1}$.

$$\begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 7-\lambda & 2 \\ -4 & 1-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 7-\lambda & 2 \\ -4 & 1-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - tr(A) + |A| = 0$$

$$\lambda^2 - 8\lambda + 15 = 0$$

$$(\lambda - 5)(\lambda - 3) = 0$$

$$\lambda = 5, 3$$

To find eigen vectors:

$$(A - \lambda I) x = 0$$

when $\lambda = 5$,

$$\begin{bmatrix} 7-5 & 2 \\ -4 & 1-5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 2x_2 = 0$$

$$-4x_1 - 4x_2 = 0$$

$x_1 = -x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} //$$

when $\lambda = 3$,

$$\begin{bmatrix} 7-3 & 2 \\ -4 & 1-3 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} +4 & 2 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$4x_3 + 2x_4 = 0$

$-4x_3 - 2x_4 = 0$

$4x_3 = -2x_4$

$-2x_3 = x_4$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_3 \\ -2x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$P = \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} //$$

$D = P^{-1} A P$

$$= \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} //$$

$A^6 = P D^6 P^{-1}$

$$= \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 5^6 & 0 \\ 0 & 3^6 \end{bmatrix}$$

$$= \begin{bmatrix} 30521 & -29 \\ 14896 & -14 \end{bmatrix}$$

2. $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

$(A - \lambda I)$

$$\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} =$$

$\lambda^2 - tr(A)\lambda + 1$

$\lambda^2 - 5\lambda + 4$

$\lambda = 4, 1$ //

When $\lambda = 4$

$$\begin{bmatrix} 3-4 & 1 \\ 2 & 2-4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$-x_1 + x_2$

$2x_1 - 2$

$$A^6 = P D^6 P^{-1}$$

$$= \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 5^6 & 0 \\ 0 & 3^6 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 30521 & -29792 \\ 14896 & -14167 \end{bmatrix} //$$

2. $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

$(A - \lambda I)$

$$\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$\lambda^2 - tr(A)\lambda + |A| = 0$

$\lambda^2 - 5\lambda + 4 = 0$

$\lambda = 4, 1$ //

When $\lambda = 4$

$$\begin{bmatrix} 3-4 & 1 \\ 2 & 2-4 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-x_1 + x_2 = 0$

$2x_1 - 2x_2 = 0 \quad \Rightarrow \quad x_1 = x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

when $\lambda = 1$,

$$\begin{bmatrix} 3-1 & 1 \\ 2 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$2x_1 + x_2 = 0$

$2x_1 + x_2 = 0$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

$D = P^{-1} A P$

$$= \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} //$$

3. $$\begin{bmatrix} -2 & 12 \\ -1 & 5 \end{bmatrix} = A$$

$A - \lambda I$

$$\begin{bmatrix} -2 & 12 \\ -1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} -2-\lambda & 12 \\ -1 & 5-\lambda \end{vmatrix} = 0$$

$\lambda^2 - tr(A)\lambda + |A|$

$\lambda^2 - 3\lambda + 2 =$

$\lambda = 2, 1$

when $\lambda = 2$,

$$\begin{bmatrix} -2-2 & 12 \\ 1 & 5-2 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 12 \\ 1 & 3 \end{bmatrix} \begin{bmatrix}$$

$-4x_1 + 12x_2$

$x_1 + 3x_2$

$x_1 =$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$$

when $\lambda = 1$

$$\begin{bmatrix} -2-1 & 1 \\ 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 12 \\ 1 & 4 \end{bmatrix}$$

$-3x_3 + 1$

$x_3 + 4$

$x_3 =$

$$\lambda^2 - tr(A)\lambda + |A| = 0$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda = 2, 1$$

$$|A| = -2(5) - (-1)(2$$
$$= -10 + 12$$
$$= 2$$

when $\lambda = 2$,

$$\begin{bmatrix} -2-2 & 12 \\ 1 & 5-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 12 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4x_1 + 12x_2 = 0$$
$$x_1 + 3x_2 = 0$$

where $x_1 = 3x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_2 \\ 1x_2 \end{bmatrix}$$

when $\lambda = 1$,

$$\begin{bmatrix} -2-1 & 12 \\ 1 & 5-1 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 12 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-3x_3 + 12x_4 = 0$$
$$x_3 + 4x_4 = 0$$

$$x_3 = 4x_4$$

$$\begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$ //

$$P = \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}$$

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -1 & 4 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} -2 & 12 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$ //

4. $\begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$

$A - \lambda I$

$$\begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} 5-\lambda & 3 \\ 1 & 3-\lambda \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 5-\lambda & 3 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$|A| = 5(3) - 1(3)$

$= 15 - 3$

$= 12$

$\lambda^2 - tr(A)\lambda + |A| = 0$

$\lambda^2 - 8\lambda + 12 = 0$

$\lambda = 6, 2$

when $\lambda = 6$,

$$\begin{bmatrix} 5-6 & 3 \\ 1 & 3- \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 \\ 1 & -3 \end{bmatrix}$$

$-x_1 + 3x_2$

$x_1 - 3$

$+x_1 = +$

$x_1 =$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$$

when λ

$$\begin{bmatrix} x_3 & 3 \\ 1 & \end{bmatrix}$$

$3x_3 + 3$

$1x_3 +$

x_3

$$\begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$$

$P = \begin{bmatrix}$

$D = P$

$$\begin{bmatrix} 5-6 & 3 \\ 1 & 3-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-x_1 + 3x_2 = 0$

$x_1 - 3x_2 = 0$

$+x_1 = +3x_2$

$x_1 = 3x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

when $\lambda = 2$

$$\begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$3x_3 + 3x_4 = 0$

$1x_3 + 1x_4 = 0$

$x_3 = -x_4$

$$\begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$

$D = P^{-1} A P$

$$= \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$$ //

5) $$\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 6 & 1 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$M_{11} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix}$ $M_{22} = \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 1 & 6 \\ 1 & 2 \end{vmatrix}$

$= 6$ $= 3$ $= -4$

$|A| = 1(6) - 6(3) + 1(0) = 12$

$\lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$

$\lambda = -1, 4, 3$

when $\lambda = -1$

$$\begin{bmatrix} 2 & 6 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x + 6y + z = 0 \quad \text{—(1)}$$

$$x + 3y = 0 \quad \text{—(2)}$$

$$4z = 0 \quad \text{—(3)}$$

$$x = -y = z = k$$

$$\begin{vmatrix} 6 & 1 \\ 0 & 4 \end{vmatrix} \quad \begin{vmatrix} 2 & 1 \\ 0 & 4 \end{vmatrix} \quad \begin{vmatrix} 2 & 6 \\ 0 & 0 \end{vmatrix}$$

$$\frac{x}{24} = \frac{-y}{8} = \frac{z}{0} = k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}$$

when $\lambda = 4$

$$\begin{bmatrix} -3 & 6 & 1 \\ 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-3x + 6y + z = 0$$

$$1x - 2y = 0$$

$$z = 0$$

$$x = -y = z = k$$

$$\begin{vmatrix} -2 & 0 \\ 0 & -1 \end{vmatrix} \quad \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} \quad \begin{vmatrix} 1 & -2 \\ 0 & 0 \end{vmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

when $\lambda = 3$,

$$\begin{bmatrix} -2 & 6 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x + 6y + z = 0$

$x - y = 0$

$$\frac{x}{\begin{vmatrix} 6 & 1 \\ -1 & 0 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & 6 \\ 1 & -1 \end{vmatrix}} = k$$

$$\frac{x}{1} = \frac{-y}{1} = \frac{z}{-4} = k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -4 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 2 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & -4 \end{bmatrix}$$

$D = P^{-1} A P$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$ //

6. $$A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$$

$\Rightarrow |A - \lambda I| = 0$

$$\left| \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$\begin{vmatrix} 6-\lambda & -2 & -1 \\ -2 & 6-\lambda & -1 \\ -1 & -1 & 5-\lambda \end{vmatrix} = 0$$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} +$

$M_{11} = \begin{vmatrix} 6 & -1 \\ -1 & 5 \end{vmatrix}$ $M_{22} =$

$= 29$

$|A| = 6(29) + 2(-11$

$= 174 - 22 -$

$= 144$

$\lambda^3 - 17\lambda^2 + 90 -$

$\lambda = 3, 6, 8$

when $\lambda = 3$,

$$\begin{bmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$3x - 2y - z =$

$-2x + 3y - z$

$-x - y + 2z$

$$\Rightarrow |A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & -1 \\ -2 & 6-\lambda & -1 \\ -1 & -1 & 5-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - tr(A)\lambda^2 + (M_{11}+M_{22}+M_{33})\lambda - |A| = 0$$

$M_{11} = \begin{vmatrix} 6 & -1 \\ -1 & 5 \end{vmatrix}$ $M_{22} = \begin{vmatrix} 6 & -1 \\ -1 & 5 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 6 & -2 \\ -2 & 6 \end{vmatrix}$

$= 29$ $= 29$ $= 32$

$$|A| = 6(29) + 2(-11) - 1(8)$$
$$= 174 - 22 - 8$$
$$= 144$$

$$\lambda^3 - 17\lambda^2 + 90 - 144 = 0$$
$$\lambda = 3, 6, 8$$

when $\lambda = 3$,

$$\begin{bmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$3x - 2y - z = 0$$
$$-2x + 3y - z = 0$$
$$-x - y + 2z = 0$$

$$\frac{x}{\begin{vmatrix} -2 & -1 \\ -1 & 2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 3 & -2 \\ -1 & -1 \end{vmatrix}} = k$$

$$\frac{x}{-5} = \frac{-y}{-5} = \frac{z}{-5} = k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} +1 \\ +1 \\ +1 \end{bmatrix}$$

when $\lambda = 6$,

$$\begin{bmatrix} 0 & -2 & -1 \\ -2 & 0 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2y - z = 0$$
$$-2x - z = 0$$
$$-x - y - z = 0$$

$$\frac{x}{\begin{vmatrix} -2 & -1 \\ -1 & -1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 0 & -1 \\ -1 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & -2 \\ -1 & -1 \end{vmatrix}} = k$$

$$\frac{x}{1} = \frac{-y}{+1} = \frac{z}{-2} = k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

when $\lambda = 8$,

$$\begin{bmatrix} -2 & -2 & -1 \\ -2 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$-2x - 2y - z = 0$$
$$-2x - 2y - z = 0$$
$$-x - y - 3z = 0$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} +1 \\ -1 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -2 & 0 \end{bmatrix}$$

$$D = P^{-1} A P$$

$$= \begin{bmatrix} 3 & 0 & \\ 0 & 6 & \\ 0 & 0 & \end{bmatrix}$$

Least squar

→ Least squar
the best ap
linear equa
exist.

→ Consider a
the appro
i.e... $(A^T A$

Working

1. Given A
2. Compute
normal

$$\begin{bmatrix} -2 & -2 & -1 \\ -2 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x - 2y - z = 0$

$-2x - 2y - z = 0$

$-x - y - 3z = 0$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} +1 \\ -1 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -2 & 0 \end{bmatrix}$$

$D = P^{-1} A P$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 8 \end{bmatrix} //$$

Least Square Method:

→ Least square solution is a method used to find the best approximate solution of system of linear equation, when an exact soln does not exist.

→ Consider a system of equation $AX = B$. To find the approximate solution we use normal equation i.e... $(A^T A)X = A^T B$ OR $X = (A^T A)^{-1}$

Working Rule:

1. Given A and B, Find A^T
2. Compute $A^T A$ and $A^T B$ and substitute in normal eqn $(A^T A)x = A^T B$

3. Solve the normal equation by any ~~AE~~ known method.

Find the Least Square sol^n of the inconsistant system $AX = B$ with

1. $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ & $B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$

$\Rightarrow A^T = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$

$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$

$A^T B = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$

$(A^T A)X = A^T B$

$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$

$17x + y = 19$

$x + 5y = 11$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} -5 \\ \end{bmatrix}$

$A^T = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix}$

$A^T A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix}$

$A^T B = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix}$

$(A^T A)X = A^T$

$\begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$12x + 8y =$

$8x + 10y$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 1 \\ 3 & - \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 1 \\ 2 & 0 \\ 2 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}$

$$A^T = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} -24 \\ -2 \end{bmatrix}$$

$$(A^T A)X = A^T B$$

$$\begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -24 \\ -2 \end{bmatrix}$$

$$12x + 8y = -24$$

$$8x + 10y = -2$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \end{bmatrix} //$$

3. $A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$ & $B = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

$$A^T = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}$$

$$A^TB = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix}$$

$$(A^TA)\,x = A^TB$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix}$$

$$3x + 3y = 6$$

$$3x + 11y = 14$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} //$$

4. $A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

$$A^T = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} +6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$A^TB = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$(A^TA)\,x = A^TB$$

$$\begin{bmatrix} +6 & -11 \\ -11 & 22 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$6x - 11y =$

$-11x + 22y =$

Q R-Factori

→ Let A be

be factori

$m \times n$ m

orthonorm

R is an

matrix.

Working

1. Let the

which f

2. Construct

from {

3. Constru

$R = Q^{-1}$

decomp

II Obtain

1. $A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$v_1 = \begin{bmatrix} \\ - \end{bmatrix}$

$$(A^TA)\,x = A^TB$$

$$\begin{bmatrix} +6 & -11 \\ -11 & 22 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$6x - 11y = -4$$
$$-11x + 22y = 11$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

QR-Factorization:

→ Let A be any matrix of order $m \times n$. A can be factorized as $A = Q \cdot R$ where Q is an $m \times n$ matrix whose columns forms an orthonormal basis for column space and R is an $n \times n$ matrix, upper triangular invertible matrix with positive entries on its diagonal

Working Rule:

1. Let the columns of matrix A be $\{v_1, v_2, \ldots v_n\}$ which form the basis of columnspace of A.
2. Construct an orthonormal basis $\{\hat{u}_1, \hat{u}_2, \ldots \hat{u}_n\}$ from $\{v_1, v_2, \ldots v_n\}$ by gramsmith method.
3. Construct $Q = [\hat{u}_1, \hat{u}_2, \ldots \hat{u}_n]$ then find $R = Q^{-1}A$ or $R = Q^TA$. The Required decomposition is $A = QR$

II Obtain QR - Factorization of the matrix

1. $A = \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ $\quad v_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\quad u_1 = v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$u_2 = v_2 - \text{Proj}_{u_1} v_2$$

$$= v_2 - \left\{ \frac{v_2 \cdot u_1}{u_1 \cdot u_1} \right\} u_1$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}}{\begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}} \right\} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \left\{ \frac{-1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1/2 \\ -1/2 \end{bmatrix}$$

$$\hat{u}_1 = \frac{u_1}{\| u_1 \|}$$

$$\| u_1 \| = \sqrt{u_1 \cdot u_1}$$

$$= \sqrt{(1) + (-1)^2}$$

$$= \sqrt{2}$$

$$\hat{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\hat{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\hat{u}_2 = \frac{u_2}{\| u_2 \|}$$

$$\| u_2 \| = \sqrt{u_2 \cdot u_2}$$

$$= \sqrt{(-1/2)^2 + (}$$

$$= 1/\sqrt{2}$$

$$\hat{u}_2 = \begin{bmatrix} -1/2 \\ -1/2 \end{bmatrix} \cdot \sqrt{2}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \hat{u}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & - \\ -1/\sqrt{2} & - \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{2} & - \\ -1/\sqrt{2} & \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \\ 0 & \end{bmatrix}$$

$$A = Q R$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix}$$

2. $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

$$\hat{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\hat{u}_2 = \frac{u_2}{\|u_2\|}$$

$$\|u_2\| = \sqrt{u_2 \cdot u_2}$$
$$= \sqrt{(-1/2)^2 + (1/2)^2}$$
$$= 1/\sqrt{2}$$

$$\hat{u}_2 = \begin{bmatrix} -1/2 \\ -1/2 \end{bmatrix} \cdot \sqrt{2} = \begin{bmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$ //

$$Q = [\ \hat{u}_1 \quad \hat{u}_2\ ]$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$R = Q^T A$$
$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix}$$ //

$$A = Q\,R$$
$$\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix}$$

2. $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u_1 = v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$u_2 = v_2 - \text{Proj}_{u_1} v_2$$

$$= v_2 - \left\{ \frac{v_2 \cdot u_1}{u_1 \cdot u_1} \right\} u_1$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}}{\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \right\} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \left\{ \frac{3}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 6/5 \\ 3/5 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix}$$

$$\hat{u}_1 = \frac{u_1}{\|u_1\|}$$

$$\|u_1\| = \sqrt{u_1 \cdot u_1} = \sqrt{(2)^2 + (1)^2} = \sqrt{5}$$

$$\hat{u}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$\hat{u}_2 = \frac{u_2}{\|u_2\|}$$

$$\|u_2\| = \sqrt{u_2 \cdot u_2}$$

$$\hat{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \end{bmatrix}$$

$$= \begin{bmatrix} 2/\sqrt{5} & - \\ 1/\sqrt{5} & \end{bmatrix}$$

$$R = Q^T \cdot A$$

$$= \begin{bmatrix} 2/\sqrt{5} & \\ -1/5\sqrt{2} & \end{bmatrix}$$

$$= \sqrt{(-1/5)^2}$$

$$= \sqrt{1/25}$$

$$= \sqrt{1/5}$$

$$= 1/\sqrt{5}$$

$$\hat{u}_2 = \sqrt{5} \begin{bmatrix} & \\ & \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \end{bmatrix}$$

$$= \begin{bmatrix} 2/\sqrt{5} & \\ 1/\sqrt{5} & \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 2/\sqrt{5} & \\ -1/\sqrt{5} & \end{bmatrix}$$

$$\hat{u}_2 = \frac{u_2}{\|u_2\|}$$

$$\|u_2\| = \sqrt{u_2 \cdot u_2} \cdot \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

$$\hat{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix} = \begin{bmatrix} -1/5\sqrt{2} \\ \sqrt{2}/5 \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \hat{u}_2 \end{bmatrix}$$

$$\begin{bmatrix} 2/\sqrt{5} & -1/5\sqrt{2} \\ 1/\sqrt{5} & \sqrt{2}/5 \end{bmatrix}$$

$$R = Q^T \cdot A$$

$$= \begin{bmatrix} 2/\sqrt{5} \\ -1/5\sqrt{2} \end{bmatrix}$$

$$= \sqrt{(-1/5)^2 + (2/5)^2}$$

$$= \sqrt{1/25 + 4/25}$$

$$= \sqrt{1/5}$$

$$= 1/\sqrt{5}$$

$$\hat{u}_2 = \sqrt{5} \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \hat{u}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

classmate

Date
Page

$$\begin{bmatrix} \sqrt{5} & 3/\sqrt{5} \\ 0 & 1/\sqrt{5} \end{bmatrix}$$

$A = QR$

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ +1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{5} & 3/\sqrt{5} \\ 0 & 1/\sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} //$$

3. Obtain QR Factorization of $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$$\Rightarrow V_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad V_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad V_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$U_1 = V_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$U_2 = V_2 - \text{Proj}_{U_1} V_2$$

$$= V_2 - \left\{ \frac{V_2 \cdot U_1}{U_1 \cdot U_1} \right\} U_1$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}} \right\} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} //$$

$$\hat{U}_1 = \frac{U_1}{\| U_1 \|}$$

$$\| U_1 \| = \sqrt{U_1 \cdot}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\hat{U}_2 = \frac{U_2}{\| U_2 \|}$$

$$\| U_2 \| =$$

$$\hat{U}_2 =$$

$$U_3 = V$$

$$\begin{bmatrix}1\\0\\1\end{bmatrix} \quad \frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix}$$

$$= \begin{bmatrix}1/2\\-1/2\\1\end{bmatrix} //$$

$$\hat{U}_1 = \frac{U_1}{\|U_1\|}$$

$$\|U_1\| = \sqrt{U_1 \cdot U_1} = \sqrt{(1)^2 + (1)^2 + 0} = \sqrt{2}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix} = \begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}$$

$$\hat{U}_2 = \frac{U_2}{\|U_2\|}$$

$$\|U_2\| = \sqrt{U_2 \cdot U_2} = \sqrt{(1/2)^2 + (-1/2)^2 + 1} = \sqrt{1/4 + 1/4 + 1}$$

$$= \sqrt{6/4}$$

$$= \sqrt{3/2}$$

$$\hat{U}_2 = \frac{\sqrt{2}}{\sqrt{3}}\begin{bmatrix}1/2\\-1/2\\1\end{bmatrix} = \begin{bmatrix}1/\sqrt{3}\sqrt{2}\\-1/\sqrt{2}\sqrt{3}\\\sqrt{2}/\sqrt{3}\end{bmatrix} = \begin{bmatrix}1/\sqrt{6}\\-1/\sqrt{6}\\\sqrt{2}/\sqrt{3}\end{bmatrix} //$$

$$U_3 = V_3 - \text{Proj}_{U_1} V_3 - \text{Proj}_{U_2} V_3$$

$$= \begin{bmatrix}0\\1\\1\end{bmatrix} - \left\{ \frac{\begin{bmatrix}0\\1\\1\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix}}{\begin{bmatrix}1\\1\\0\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix}} \right\} \begin{bmatrix}1\\1\\0\end{bmatrix} =$$

$$\left\{ \frac{\begin{bmatrix}0\\1\\1\end{bmatrix}\begin{bmatrix}1/2\\-1/2\\1\end{bmatrix}}{\begin{bmatrix}1/2\\-1/2\\1\end{bmatrix}\begin{bmatrix}1/2\\-1/2\\1\end{bmatrix}} \right\} \begin{bmatrix}1/2\\-1/2\\1\end{bmatrix}$$

$$= \begin{bmatrix}0\\1\\1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix} - \begin{bmatrix}1/2\\-1/2\\1\end{bmatrix}\frac{1/2}{3/2}$$

$$= \begin{bmatrix}0\\1\\1\end{bmatrix} - \begin{bmatrix}-1/2\\-1/2\\0\end{bmatrix} - \begin{bmatrix}1/6\\-1/6\\1/3\end{bmatrix}$$

$$= \begin{bmatrix}-2/3\\2/3\\+2/3\end{bmatrix}$$

$$\hat{U}_3 = \frac{U_3}{\|U_3\|}$$

$$\|U_3\| = \sqrt{U_3 \cdot U_3} = \sqrt{(-2/3)+(2/3)+(2/3)^2} = \sqrt{4/9+4/9+4/9}$$

$$= \sqrt{4/9\times 3} = 2/\sqrt{3} //$$

$$\hat{U}_3 = \frac{\sqrt{3}}{2}\begin{bmatrix}-2/3\\2/3\\2/3\end{bmatrix} = \begin{bmatrix}$$

$$Q = \begin{bmatrix}\hat{U}_1 & \hat{U}_2 & \hat{U}_3\end{bmatrix}$$

$$= \begin{bmatrix}1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3}\\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3}\\ 0 & \sqrt{3}/\sqrt{3} & 1/\sqrt{3}\end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix}1/\sqrt{3} & 1/\sqrt{2} & 0\\ 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{3}\\ -1/\sqrt{3} & 1/\sqrt{3} & \end{bmatrix}$$

$$= \begin{bmatrix}\sqrt{2} & 1/\sqrt{2} & \\ 0 & \sqrt{3}/\sqrt{2} & \\ 0 & 0 & \end{bmatrix}$$

$$A = QR$$

$$= \begin{bmatrix}1/\sqrt{2} & 1/\sqrt{6} & \\ 1/\sqrt{2} & -1/ & \\ 0 & \sqrt{} & \end{bmatrix}$$

$$= \begin{bmatrix}1 & \\ 1 & \\ 0 & \end{bmatrix}$$

4. $\begin{bmatrix}2 & 0\\ 0 & 1\\ & 1\end{bmatrix}$

$\frac{1}{\sqrt{6}} \cdot + \frac{\sqrt{2}}{\sqrt{3}}$

$$\hat{U}_2 = \frac{\sqrt{3}}{2}\begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} - \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{U}_{\parallel} & \hat{U}_{\perp} & \hat{U}_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{12} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{12} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & \sqrt{3}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 0 \\ 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{2}/\sqrt{3} \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3}/\sqrt{2} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

$$A = QR$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & \sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3}/\sqrt{2} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Q. $\begin{bmatrix} 2 & 0 & 3 & 1 \\ 0 & 1 & 6 & 2 \\ & & 0 & 2 \end{bmatrix}$

$\frac{1}{2}$ +

$$\Rightarrow V_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \qquad V_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$U_1 = V_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$U_2 = V_2 - \text{Proj}_{U_1} V_2$$

$$= V_2 - \left\{ \frac{V_2 \cdot U_1}{U_1 \cdot U_1} \right\} U_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}}{\begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}} \right\} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\hat{U}_2 = \frac{U_2}{\|U_2\|}$$

$$\|U_2\| = \sqrt{U_2 \cdot U_2}$$

$$\hat{U}_2 = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\hat{U}_1 = \frac{U_1}{\|U_1\|}$$

$$\|U_1\| = \sqrt{U_1}$$

$$\hat{U}_1 = \frac{1}{3\sqrt{5}}$$

$$Q = \begin{bmatrix} \hat{U} \end{bmatrix}$$

$$= \begin{bmatrix} \end{bmatrix}$$

$$R = Q$$

$$= \begin{bmatrix} \end{bmatrix}$$

$$2 \times 3\sqrt{5} + 2$$

$$\|u_2\| = \sqrt{u_2 \cdot u_2} = \sqrt{(2)^2} = 2$$

$$\hat{u}_2 = \frac{1}{2}\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\hat{u}_1 = \frac{u_1}{\|u_1\|}$$

$$\|u_1\| = \sqrt{u_1 \cdot u_1} = \sqrt{(3)^2 + (6)^2} = 3\sqrt{5}$$

$$\hat{u}_1 = \frac{1}{3\sqrt{5}}\begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} \hat{u}_1 & \hat{u}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{5} & 0 \\ 2/\sqrt{5} & 0 \\ 0 & 1 \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 15/\sqrt{5} & 5/\sqrt{5} \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 15/\sqrt{5} & \sqrt{5} \\ 0 & 2 \end{bmatrix}$$

$\therefore A = QR$ //

Quadratic form of matrix:

→ A Quadratic form of a matrix is a scalar value function $QX = X^T AX$ where $A - n \times n$ symmetric matrix & x - column vector in $\mathbb{R}^n$.

→ The matrix A w.r.t 3 variables x, y, z is given as follows:

$$A = \begin{bmatrix} \text{Coef}(x^2) & \frac{1}{2}\text{Coef}(xy) & \frac{1}{2}\text{Coef}(xz) \\ \frac{1}{2}\text{Coef}(xy) & \text{Coef}(y^2) & \frac{1}{2}\text{Coef}(yz) \\ \frac{1}{2}\text{Coef}(xz) & \frac{1}{2}\text{Coef}(yz & \text{Coef}(z^2) \end{bmatrix}$$

→ if x represents variable vector in $\mathbb{R}^n$ then change of variable is an equation of the form $x = Py$ where P - invertible matrix and y - new variable vector in $\mathbb{R}^n$

→ If the change of variable made in quadratic form we get $x^T Ax = (Py)^T A(Py)$

$$= y^T (P^T AP) y$$

$$= y^T D y$$

where $D = P^T AP$ is a diagonal matrix with eigen values as a diagonal elements.

Working Rule

1. For a given quadratic form, obtain the matrix A of the same.
2. ~~Take~~ Find the eigen values of A & substitute in the canonical form.

I Reduce the follo
canonical form

1. $x_1^2 - 8x_1x_2 - 5$

$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$

$|A - \lambda I| = 0$

$\begin{vmatrix} 1-\lambda & -4 \\ -4 & -5-\lambda \end{vmatrix}$

$\lambda^2 - tr(A)\lambda +$

$\lambda^2 - 4\lambda - 21$

$(\lambda - 3)(\lambda +$

$\lambda = -7, 3$

$\therefore$ Cannonica

$\lambda_1 x$

$= 3Y$

2. $3x_1^2 - 4x$

$A = \begin{bmatrix} 3 & \\ -2 & \end{bmatrix}$

$|A - \lambda I$

$\begin{vmatrix} 3-\lambda & \\ -2 & \end{vmatrix}$

λ^2

λ^2

I. Reduce the following quadratic form into canonical form

1. $x_1^2 - 8x_1x_2 - 5x_2^2$

$$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & -4 \\ -4 & -5-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - tr(A)\lambda + |A| = 0$$

$$\lambda^2 - 4\lambda - 21 = 0$$

$$(\lambda - 3)(\lambda + 7) = 0$$

$$\lambda = -7, 3$$

∴ Cannonical form is

$$\lambda_1 Y_1^2 + \lambda_2 Y_2^2$$

$$= 3Y_1^2 - 7Y_2^2 //$$

2. $3x_1^2 - 4x_1x_2 + 6x_2^2$

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - tr(A)\lambda + |A| = 0$$

$$\lambda^2 - 9\lambda - 14 = 0$$

$3(6) - (-2)(-2)$

$18 - 4$

$\lambda = 7, 2$

$\therefore \lambda_1 Y_1^2 + \lambda_2 Y_2^2 = 2Y_1^2 + 7Y_2^2$ //

3. $2x_1^2 + 10x_1x_2 + 2x_2^2$

$$A = \begin{bmatrix} 2 & 5 \\ 5 & 2 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 5 \\ 5 & 2-\lambda \end{vmatrix} = 0$$

$\lambda^2 - tr(A)\lambda + |A| = 0$

$\lambda^2 - 4\lambda - 21 = 0$

$\lambda = 7, -3$

$\therefore -3Y_1^2 + 7Y_2^2$

4. $x_1^2 - 6x_1x_2 + 9x_2^2$

$$\Rightarrow A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & -3 \\ -3 & 9-\lambda \end{vmatrix} = 0$$

$\lambda^2 - tr(A)\lambda + |A| = 0$

$\lambda^2 - 10\lambda + 0 = 0$

$\lambda = 10, 0$

$|A| = 2(2) - 5(5$

$= 4 - 25$

$= -21$

$\therefore 0Y_1^2 + 10Y_2^2$ //

NOTE:

→ The Quadratic f
definite if a

→ -ve definite

→ Indefinite if

II Check whethe
+ve definite

1 $3x_1^2 + 2x_2^2$

$\Rightarrow A = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

$|A - \lambda I| =$

$\begin{vmatrix} 3-\lambda & \\ 2 & 2 \\ 0 & \end{vmatrix}$

$\lambda^3 - tr($

$|A| =$

$M_{11} =$

$=$

λ^3

$\therefore \; 0Y_1^2 + 10Y_2^2$ //

NOTE:

→ The Quadratic form Q(x) is said to be +ve definite if all $\lambda i > 0$ or $Q(x) > 0 \; \forall \; x \neq 0$

→ -ve definite if $\forall \lambda i < 0$ or $Q(x) < 0 \; \forall \; x \neq 0$

→ Indefinite if it has both +ve & -ve eigen values

II Check whether the following quadratic forms are +ve definite, -ve definte or indefinite.

1 $3x_1^2 + 2x_2^2 + 3x_3^2 + 4x_1x_2 + 4x_2x_3$

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 3 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 2 & 0 \\ 2 & 2-\lambda & 2 \\ 0 & 2 & 3-\lambda \end{vmatrix} = 0$$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33}) - |A| = 0$

$|A| = 3(6-4) - 2(6) = 6 - 12 = -6$ //

$M_{11} = \begin{vmatrix} 2 & 2 \\ 2 & 3 \end{vmatrix}$ $M_{22} = \begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix}$

$= 2$ $= 9$ $= 2$

$\lambda^3 - 8\lambda^2 + 13\lambda + 6 = 0$

∴ Indefinite (Given Quadratic form)

2. $3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_1x_2 + 2x_1x_3 - 2x_2x_3$

$$\Rightarrow A = \begin{bmatrix} 3 & -1 & +1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$M_{11} = \begin{vmatrix} 5 & -1 \\ -1 & 3 \end{vmatrix}$ $M_{22} = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 3 & -1 \\ -1 & 5 \end{vmatrix}$

$= 14$ $= 8$ $= 14$

$|A| = 3(15-1) + 1(-3+1) + 1(1-5)$

$= 42 - 2 - 4$

$= 36$

$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$

$\lambda = 6, 3, 2$

∴ Given quadratic form is +ve definite

3. $2x_1x_2 + 2x_1x_3 - 2x_2x_3$

⇒

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{vmatrix}$$

$\lambda^3 - tr(A)\lambda$

$M_{11} = \begin{vmatrix} 0 & - \\ -1 & \end{vmatrix}$

$= -1$

$|A| = 0 -$

$\Rightarrow \lambda^3 - 0$

$\lambda = -2$

∴ Given

4. $-x_1^2 -$

$\Rightarrow A =$

$|A - \lambda$

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$M_{11} = \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix}$ $M_{22} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$

$= -1$ $= -1$ $= -1$

$$|A| = 0 - 1(1) + 1(-1) = -1 - 1 = -2$$

$$\Rightarrow \lambda^3 - 0\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda = -2, 1, -1$$

∴ Given quadratic form is indefinite.

4. $-x_1^2 - 2x_2^2 - 3x_3^2 + 2x_1x_2$

$$\Rightarrow A = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} -1-\lambda & 1 & 0 \\ 1 & -2-\lambda & 0 \\ 0 & 0 & -3-\lambda \end{vmatrix} = 0$$

1 $\lambda^3 - tr(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$

$M_{11} = \begin{vmatrix} -2 & 0 \\ 0 & -3 \end{vmatrix}$ $M_{22} = \begin{vmatrix} -1 & 0 \\ 0 & -3 \end{vmatrix}$ $M_{33} = \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix}$

$= +6$ $= 3$ $= 1$

$|A| = -1(6) - 1(-3) = -3$

$\Rightarrow \lambda^3 + 6\lambda^2 + 10\lambda + 3 = 0$

$\therefore$ Quadratic form is -ve definite.

Singular Value Decomposition of a matrix: